

## Research Experience

- 09/2026– **Postdoctoral Associate**, *Medical Image Processing and Machine Learning Laboratory (MIPLab)*, Calgary, Canada, Department of Radiology.  
Implement and evaluate vision-language model-based digital twins of Alzheimer's disease.
- 02/2018– **Student Research Assistant**, *Fraunhofer Institute for Algorithms and Scientific Computing (SCAI)*, Sankt Augustin, Germany, Department of Bioinformatics.  
10/2021 Implemented natural language processing methods in biomedical contexts and developed machine learning approaches for Alzheimer's disease diagnosis.

## Education

- 09/2025– **Visiting PhD Student, Prof. Fyshe's Lab**, *University of Alberta & Alberta Machine Intelligence Institute (Amii)*, Edmonton, Canada, Department of Computing Science.  
08/2026 Vision-Language Models, Compositionality, Neural Encoding, Knowledge Graph Representation Learning, Functional Magnetic Resonance Imaging (MRI)
- 11/2021– **Interdisciplinary PhD Student & Fonds Wetenschappelijk Onderzoek (FWO) Fellow**, *KU Leuven*, Leuven, Belgium, Departments of Neurosciences and Computer Science.  
06/2026 Natural Language Processing, Neurolinguistics, Neural Encoding, Alzheimer's Disease, Connected Speech, Functional Magnetic Resonance Imaging (MRI)
- 09/2018– **Master of Science in Computer Science**, *Bonn-Rhein-Sieg University of Applied Sciences*,  
08/2021 Sankt Augustin, Germany.  
Natural Language Processing, Bioinformatics, Alzheimer's Disease, Knowledge Graphs
- 09/2015– **Bachelor of Science in Computer Science**, *Bonn-Rhein-Sieg University of Applied Sciences*,  
08/2018 Sankt Augustin, Germany.  
Bioinformatics, Software Engineering, Machine Learning
- 09/2017– **Visiting Undergraduate Student, Prof. Beiko's Lab**, *Dalhousie University*, Halifax, Canada,  
12/2017 Department of Computer Science.  
Metagenomics, Bioinformatics, Machine Learning

## Teaching Experience

- 09/2018– **Teaching Assistant**, *Bonn-Rhein-Sieg University of Applied Sciences*, Sankt Augustin, Germany,  
01/2020 Introduction to Probability Theory and Statistics.  
Led exercise sessions for 60+ students per year covering Python programming and core statistics concepts.
- 09/2016– **Tutor**, *Bonn-Rhein-Sieg University of Applied Sciences*, Sankt Augustin, Germany, Introductory  
06/2019 Computer Science Courses.  
Assisted students with assignments and exam preparation across multiple introductory courses.

## Academic Service

- Reviewing npj Digital Medicine, Alzheimer's Research & Therapy, Bioinformatics, Journal of Neurology,  
Activities Bioinformatics Advances, Scientific Reports, BMC Medical Informatics & Decision Making, npj Dementia, International Journal of Language & Communication Disorders, European Conference on Artificial Intelligence (ECAI)

Master's Thesis Supervision Ella Eycken – Combining Logopedics and Natural Language Processing for Neurodegeneration.  
Ali Tolga Dincer – Parameter-Efficient Fine-Tuning in Large Language Models for Brain Decoding.

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## Invited Talks

- 03/2025 **Natural Language Processing in the Human Brain**, *CLIPS Colloquium*, University of Antwerp, Antwerp, Belgium.
- 01/2026 **Language Models in Alzheimer's Disease**, *CURE-ND Early Career Researcher Session*, World Brain Health Forum, Paris, France.

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## Awards

- 03/2024 Best Oral Presentation – CURE-ND Early Researchers Workshop
- 11/2022–08/2026 PhD Fellowship Fundamental Research – Research Foundation Flanders (FWO) (120,000€ over the course of 4 years)
- 06/2022 Studienpreis – Anwenderforum Bonn e.V. (5,000€)
- 09/2016–08/2019 Deutschlandstipendium – Bonn-Rhein-Sieg University of Applied Sciences (300€/month)
- 10/2018 Best Thesis Award – Bonn-Rhein-Sieg University of Applied Sciences (1,000€)
- 06/2017–09/2017 International Study and Training Partnerships scholarship – German Academic Exchange Service/Bonn-Rhein-Sieg University of Applied Sciences (800€/month)

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## Publications

For an up-to-date list of all publications, please see my google scholar profile.

- 06/2026 Compositional Complexity in Text and Images. *Neurobiology of Language*. **Helena Balabin**, Antonietta Gabriella Liuzzi, Kevin Statz, Marie-Francine Moens, Patrick Dupont, Rik Vandenberghe.
- 06/2025 Evaluating Language Model Embeddings for Parkinson's Disease Cohort Harmonization Using a Novel Manually Curated Variable Mapping Schema. *Scientific Reports*. Yasamin Salimi, Tim Adams, Mehmet Can Ay, **Helena Balabin**, Marc Jacobs, Martin Hofmann-Apitius.
- 01/2025 Natural Language Processing-Based Classification of Early Alzheimer's Disease from Connected Speech. *Alzheimer's & Dementia*. **Helena Balabin**, Bastiaan Tamm, Laure Spruyt, Nathalie Dusart, Ines Kabouche, Ella Eycken, Kevin Statz, Steffi De Meyer, Hugo Van Hamme, Patrick Dupont, Marie-Francine Moens, Rik Vandenberghe.
- 06/2024 Semantic Harmonization of Alzheimer's Disease Datasets Using AD-Mapper. *Journal of Alzheimer's Disease*. Philipp Wegner, **Helena Balabin**, Mehmet Can Ay, Sarah Bauermeister, Lewis Killin, John Gallacher, Martin Hofmann-Apitius, Yasamin Salimi.
- 02/2024 A Comparative Analysis of Language Models for the Classification of Alzheimer's Disease Based on Connected Speech. *AAAI Workshop on Artificial Intelligence for Brain Encoding and Decoding*. **Helena Balabin**, Laure Spruyt, Ella Eycken, Ines Kabouche, Bastiaan Tamm, Jolien Schaefferbeke, Patrick Dupont, Marie-Francine Moens, Rik Vandenberghe.
- 10/2023 Investigating Neural Fit Approaches for Sentence Embedding Model Paradigms. *ECAI*. **Helena Balabin**, Antonietta Gabriella Liuzzi, Jingyuan Sun, Patrick Dupont, Rik Vandenberghe, Marie-Francine Moens.
- 10/2022 How Relevant Is Selective Memory Population in Lifelong Language Learning? *AACL/IJCNLP*. Vladimir Araujo, **Helena Balabin**, Julio Hurtado, Alvaro Soto, Marie-Francine Moens.

- 10/2022 Pre-trained Speech Representations as Feature Extractors for Speech Quality Assessment in Online Conferencing Applications. *Interspeech*. Bastiaan Tamm, **Helena Balabin**, Rik Vandenberghe, Hugo Van Hamme.
- 05/2022 Deep Learning-based Detection of Psychiatric Attributes from German Mental Health Records. *International Journal of Medical Informatics*. Sumit Madan, Fabian Julius Zimmer, **Helena Balabin**, Sebastian Schaaf, Holger Froehlich, Juliane Fluck, Irene Neuner, Klaus Mathiak, Martin Hofmann-Apitius, Pegah Sarkheil.
- 03/2022 STonKGs: A Sophisticated Transformer Trained on Biomedical Text and Knowledge Graphs. *Bioinformatics*. **Helena Balabin**, Charles Tapley Hoyt, Colin Birkenbihl, Benjamin M Gyori, John Bachman, Alpha Tom Kodamullil, Paul G Ploeger, Martin Hofmann-Apitius, Daniel Domingo-Fernandez.
- 01/2022 ProtSTonKGs: A Sophisticated Transformer Trained on Protein Sequences, Text, and Knowledge Graphs. *SWAT4HCLS*. **Helena Balabin**, Charles Tapley Hoyt, Benjamin M Gyori, John Bachman, Alpha Tom Kodamullil, Martin Hofmann-Apitius, Daniel Domingo-Fernandez.
- 12/2020 The COVID-19 Ontology. *Bioinformatics*. Astghik Sargsyan, Alpha Tom Kodamullil, Shounak Baksi, Johannes Darms, Sumit Madan, Stephan Gebel, Oliver Keminer, Geena Mariya Jose, **Helena Balabin**, Lauren Nicole DeLong, Manfred Kohler, Marc Jacobs, Martin Hofmann-Apitius.

## Skills

Technical Python (huggingface, pytorch, scikit-learn); MLOps (git, docker, mlflow, pytest, hydra); fMRI scanning & processing (BIDS, psychopy, fMRIPrep, GLMSingle); Linux (HPC, slurm, remote workflows)

## Languages

German Native  
 English Fluent  
 Dutch Fluent  
 Russian Conversational

## Interests

Artificial Intelligence Natural Language Processing, Compositionality, Vision-Language Models, Knowledge Graphs, Neural En-/Decoding  
 Neuroscience Neurolinguistics, Compositionality, Cognitive Neurology, Alzheimer's Disease, Functional MRI